Binance Gold & Silver Perpetuals and ETF Markets

KSE Python course project

1. Introduction and research question

Research question.
Did the introduction of Binance TRADFI gold and silver perpetual futures change volatility, liquidity and price integration in the corresponding precious metal ETF markets (GLD and SLV)?

Main hypotheses (testable with pre/post data).

- **H1 (volatility)**: Realized volatility of gold and silver ETFs increased after the launch of Binance XAUUSDT and XAGUSDT TRADFI perpetual futures.
- **H2 (liquidity)**: Trading activity and liquidity of GLD and SLV improved (higher volume, lower Amihud illiquidity) after the launch.

We do *not* formulate a formal hypothesis on "integration increased", because Binance returns do not exist before the launch, so we cannot compute pre-event GLD–XAU or SLV–XAG correlation or beta. We still report post-event rolling correlation and integration metrics as **descriptive** evidence (how strongly ETF and perpetual co-move after the launch), but we do not test a pre/post change for integration.

How we identify the impact on the market. We use an **event study** with (i) pre/post comparison of means and t-tests, and (ii) **OLS regressions** where the outcome is volatility (or liquidity proxy) and the key regressor is a **post-event dummy** (`post_event` = 1 after the launch date), plus controls (DXY, VIX, 10Y yield changes). The coefficient on `post_event` is our estimate of the change in the outcome after the launch, conditional on controls. We do not have a control group (e.g. other assets without a new derivative), so we do not use difference-in-differences; the design is a single-market before/after event study with controls. See `Literature/LITERATURE_METHODS.md` for a comparison with the Bitcoin futures paper (Augustin et al., Management Science 2023), which uses DiD.

2. Data and feature set

We use three pre-built datasets produced by `build_features.py`:

- **Daily panel** (`features_daily.csv`): 118 trading days, 87 variables. Used for the event study (pre/post comparison, regressions).
- **Hourly panel** (`features_hourly.csv`): one row per hour over the same calendar span; Binance 1m aggregated to hourly, ETF/controls forward-filled from daily.
- **1-minute panel** (`features_1m.csv`): one row per Binance XAUUSDT 1m candle with daily ETF/controls merged by date. Used to satisfy the course requirement of **10,000 observations**.

Raw data sources and variable definitions are in `DATA_DESCRIPTION.md`, `VARIABLES_DESCRIPTION.md`, and `DATA_ROW_COUNTS_AND_PERIODS.md`. A mapping of which variables are used where (and how to extend usage) is in `VARIABLE_USAGE.md`.

```
Daily: 118 rows (features_daily.csv)
Hourly: 4104 rows (features_hourly.csv)
1-minute: 67074 rows (features_1m.csv)
```

```
(118, 89)
```

First rows of the daily panel (first 8 of 87 columns shown so the table fits on the page in PDF).

Date	px_gld_open	px_gld_high	px_gld_low	px_gld_close	vol_gld	px_slv_open	px_slv_high	px_
2025-09-02	320.820007	325.920013	320.239990	325.589996	21247500.0	36.630001	37.169998	36.4
2025-09-03	327.760010	329.450012	326.730011	328.140015	16062200.0	37.330002	37.639999	37.1
2025-09-04	327.029999	327.559998	325.350006	326.690002	11703100.0	37.200001	37.220001	36.6
2025-09-05	329.529999	331.440002	328.929993	331.049988	16065200.0	37.410000	37.599998	36.9
2025-09-08	333.660004	335.670013	333.230011	334.820007	18617300.0	37.570000	37.830002	37.3

2.1 Basic cleaning and event date

We define the **event** as the **launch** of Binance XAUUSDT TRADFI perpetual futures. So there is one date: 2026-01-05 (event = launch).
To keep a meaningful pre-event window, we only require that ETF prices and returns are available; Binance-specific fields are allowed to be missing before the launch.

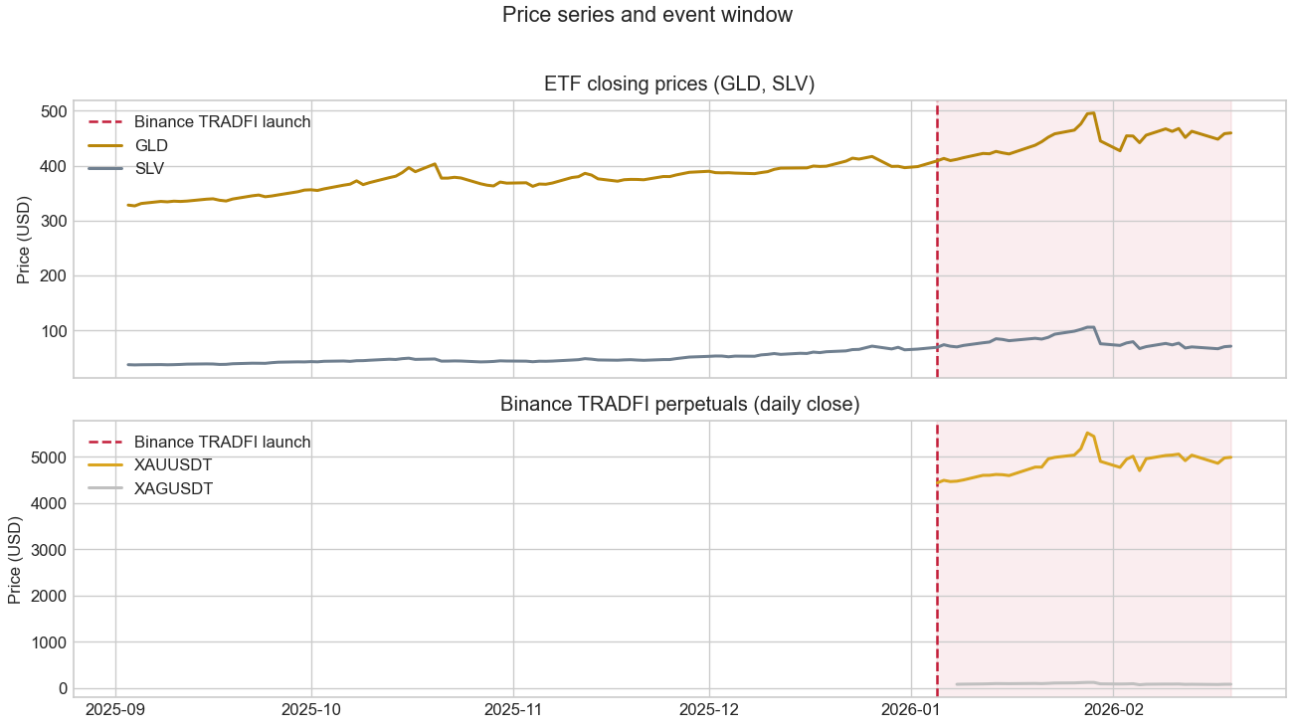
```
Range: 2025-09-03 to 2026-02-19, 117 rows × 89 columns.
```

```
px_gld_open     float64
px_gld_high     float64
px_gld_low      float64
px_gld_close    float64
vol_gld         float64
px_slv_open     float64
px_slv_high     float64
px_slv_low      float64
px_slv_close    float64
vol_slv         float64
dxy             float64
vix             float64
dtype: object
```

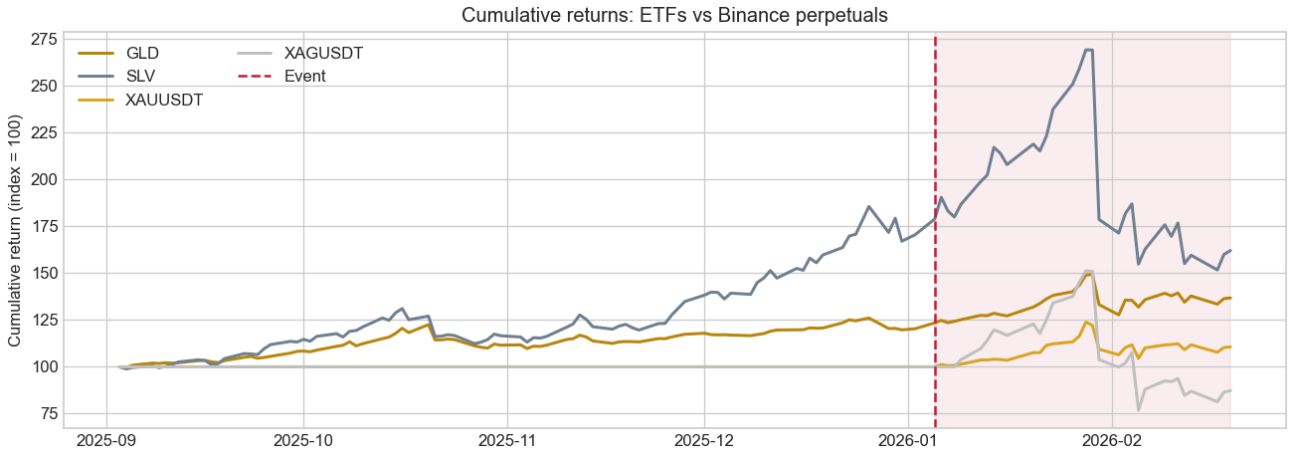
```
post_event
0             85
1             32
Name: count, dtype: int64
```

3. Descriptive statistics

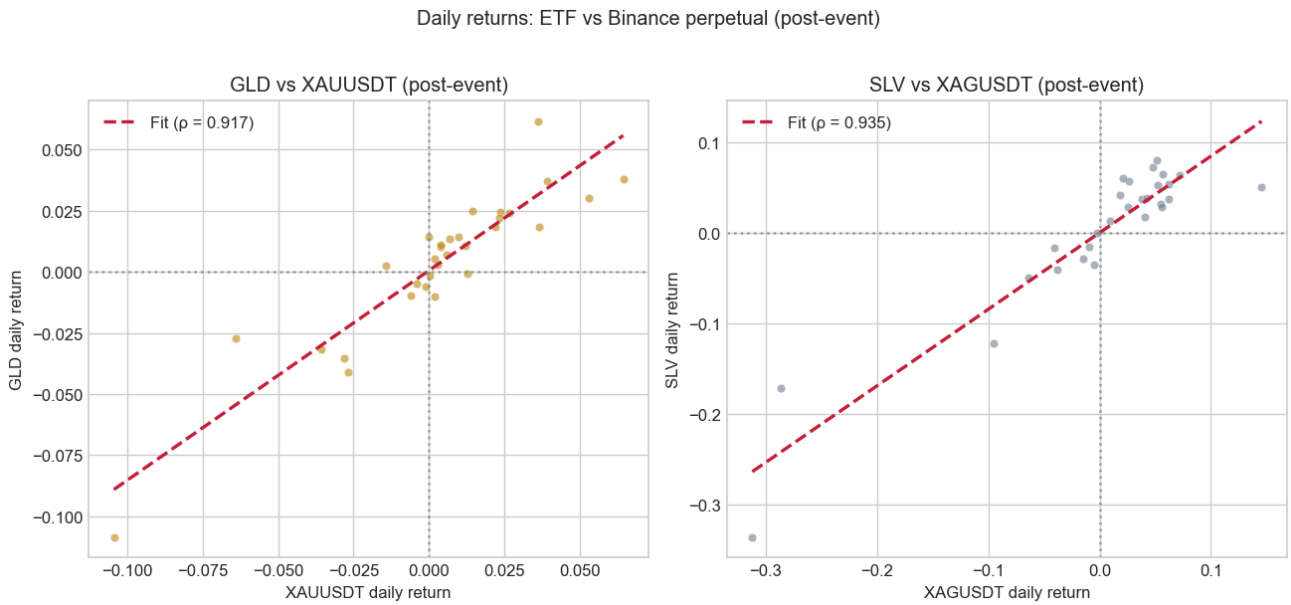
3.1 Prices and returns



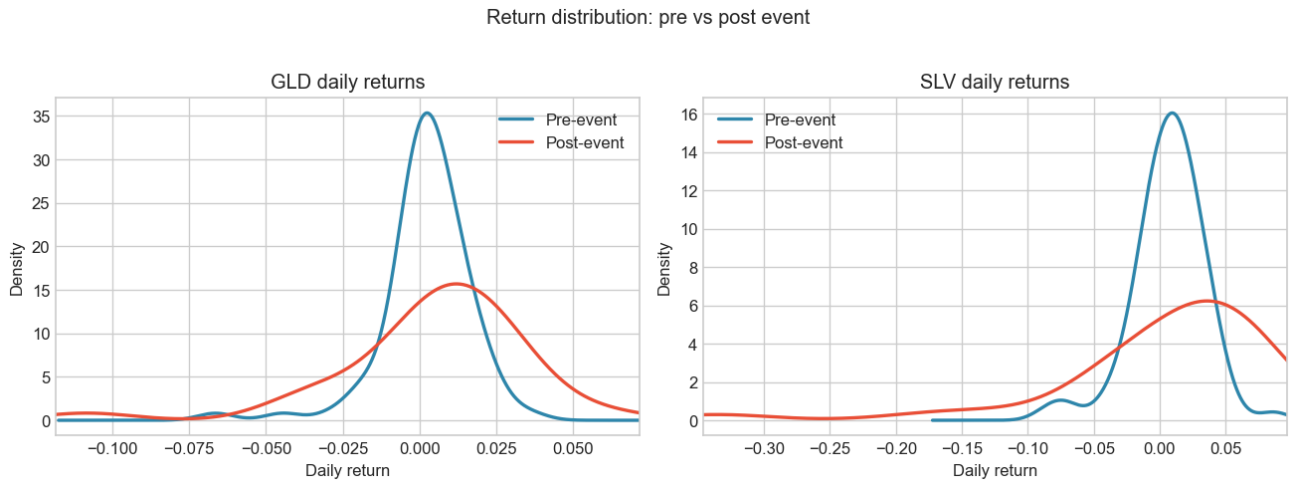
Cumulative returns (normalized to 100 at the first date): a clear view of relative performance and whether paths diverge after the event.



Post-event integration in one picture: scatter of ETF daily returns vs Binance perpetual returns (post-event only). Points along the diagonal mean the two move together; the regression line and correlation summarize the link.



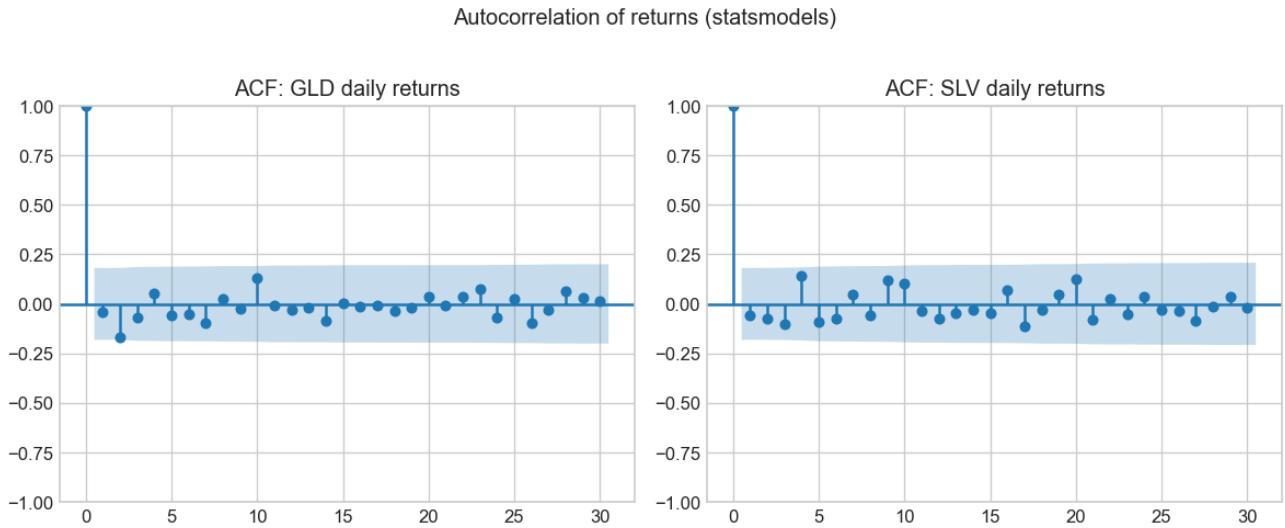
How the distribution of returns changed: pre- vs post-event density of daily returns. A wider (more dispersed) distribution after the event is consistent with higher volatility.



Python analytics: interactive visualization (Plotly) — as in the class examples (e.g. Debt vs Renew), we use Plotly for one interactive chart: hover over dates to see values.

```
Plotly not available: No module named 'plotly'
```

Time-series diagnostics (statsmodels): ACF of daily returns to show short-run dependence, as in the class time-series examples (e.g. AutoReg/VAR project).

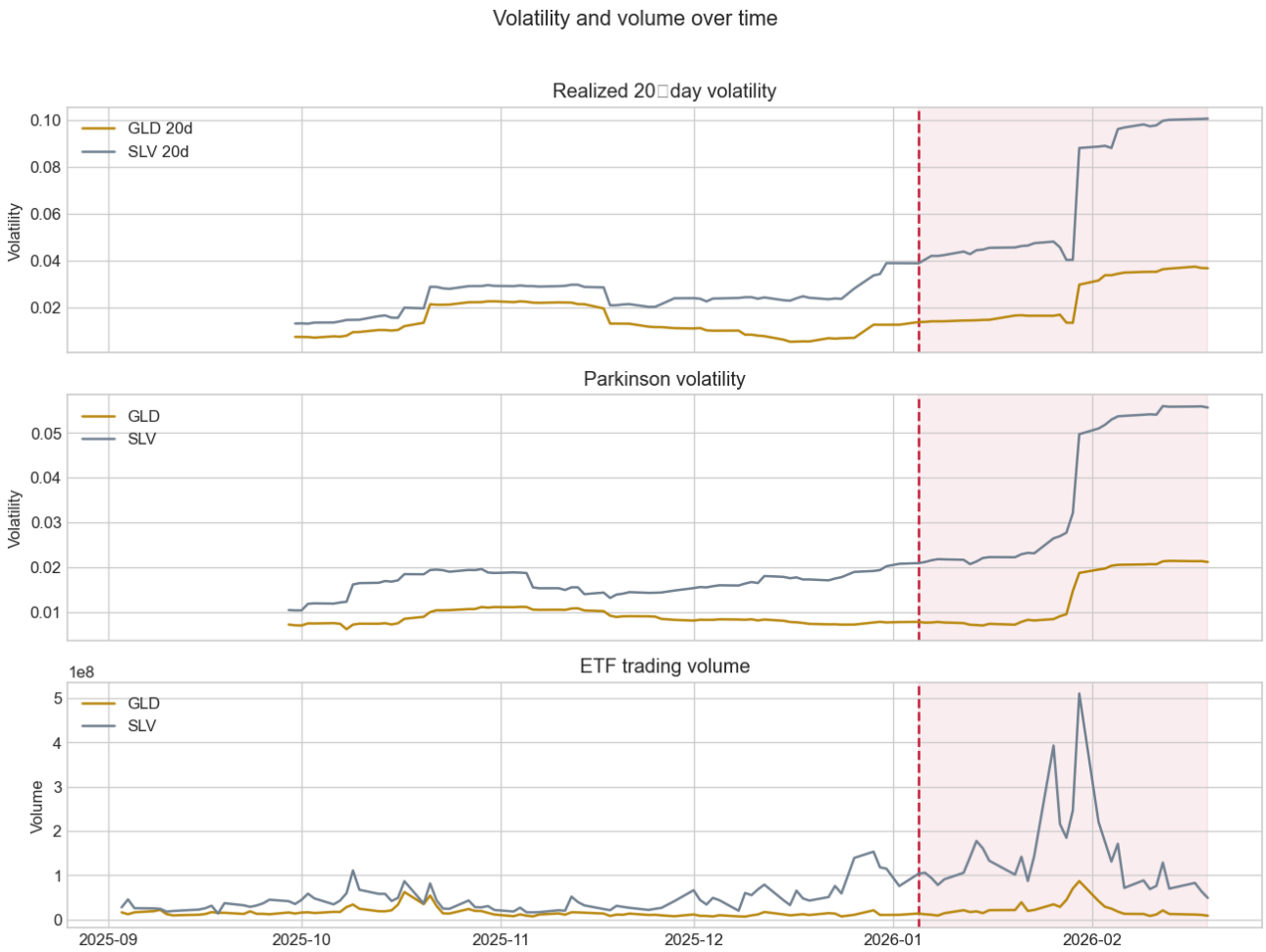


A concise list of Python techniques used in this project (pandas, numpy, scipy, statsmodels, Plotly, etc.), aligned with the course References, is in `PYTHON_TECHNIQUES.md`.

	mean	std	min	max
ret_gld	0.002946	0.019728	-0.108412	0.061647
ret_slv	0.005537	0.049015	-0.336037	0.086602
ret_xau_binance	0.003788	0.032585	-0.104457	0.064529
ret_xag_binance	0.000494	0.097086	-0.312222	0.145629

3.2 Volatility and liquidity over time

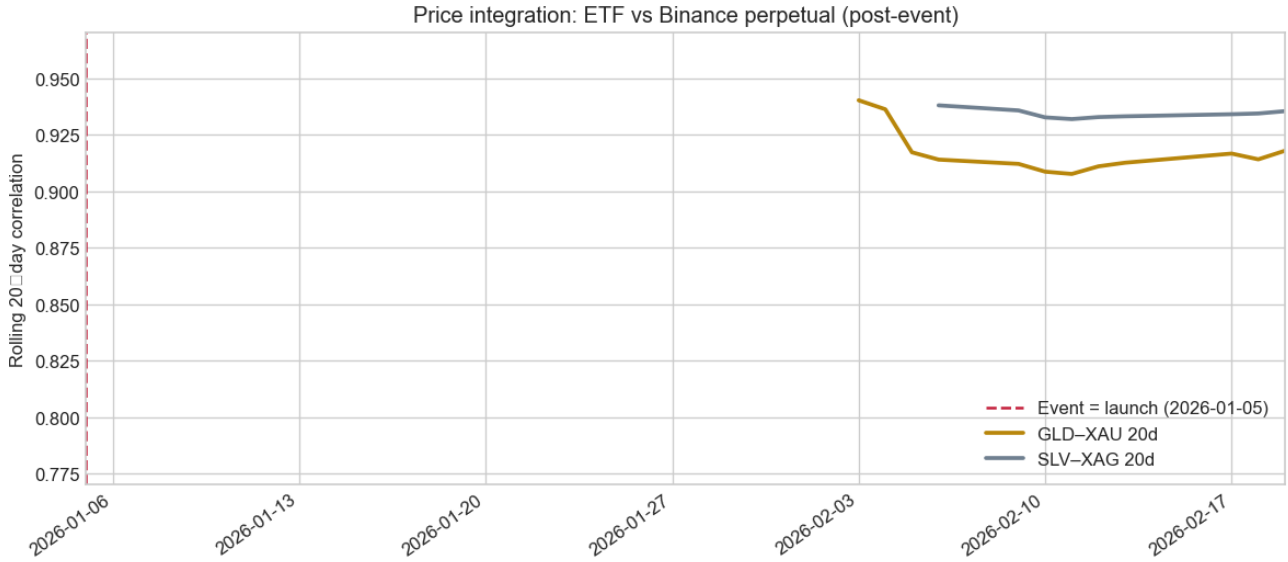
We look at 20-day realized volatility for GLD and SLV and ETF trading volume.



Why there is no pre-period for integration. The event is the *launch* of Binance XAUUSDT/XAGUSDT. Before that date there are no Binance prices or returns, so series like `corr_20_gld_xau` and `beta_60_gld_on_xau` (which need both ETF and Binance returns) are undefined before the launch. We therefore **cannot** estimate "integration before the event" in the same way we do for volatility or liquidity: there is no pre-period for Binance–ETF co-movement. We can only describe integration *after* the launch (level and evolution of rolling correlation) and, in the pre/post table, the "pre" mean for correlation is effectively missing or based on no data.

What this shows: The *rolling 20-day correlation* is the correlation between daily returns of the ETF (GLD or SLV) and the Binance perpetual (XAUUSDT or XAGUSDT) over the last 20 days. Values near 1 mean the two series move together. This metric exists only when both sides have data (post-event). The chart and the pre/post table in section 4 show post-event integration only; there is no comparable pre-event integration measure.

Single chart (gold and silver together): Same scale and dates so both lines are easy to compare. Y-axis is limited to the range where the data lies so the lines are not squeezed at the top. The vertical line is the **event date** — that is, the **launch date** of Binance XAUUSDT (2026-01-05). We use this single date as "the event" everywhere in the report. The SLV–XAG line starts later than GLD–XAU because XAGUSDT was listed a few days after XAUUSDT, so the 20-day rolling window has enough data only from a later date.



Minute-level correlation (last 7 days)

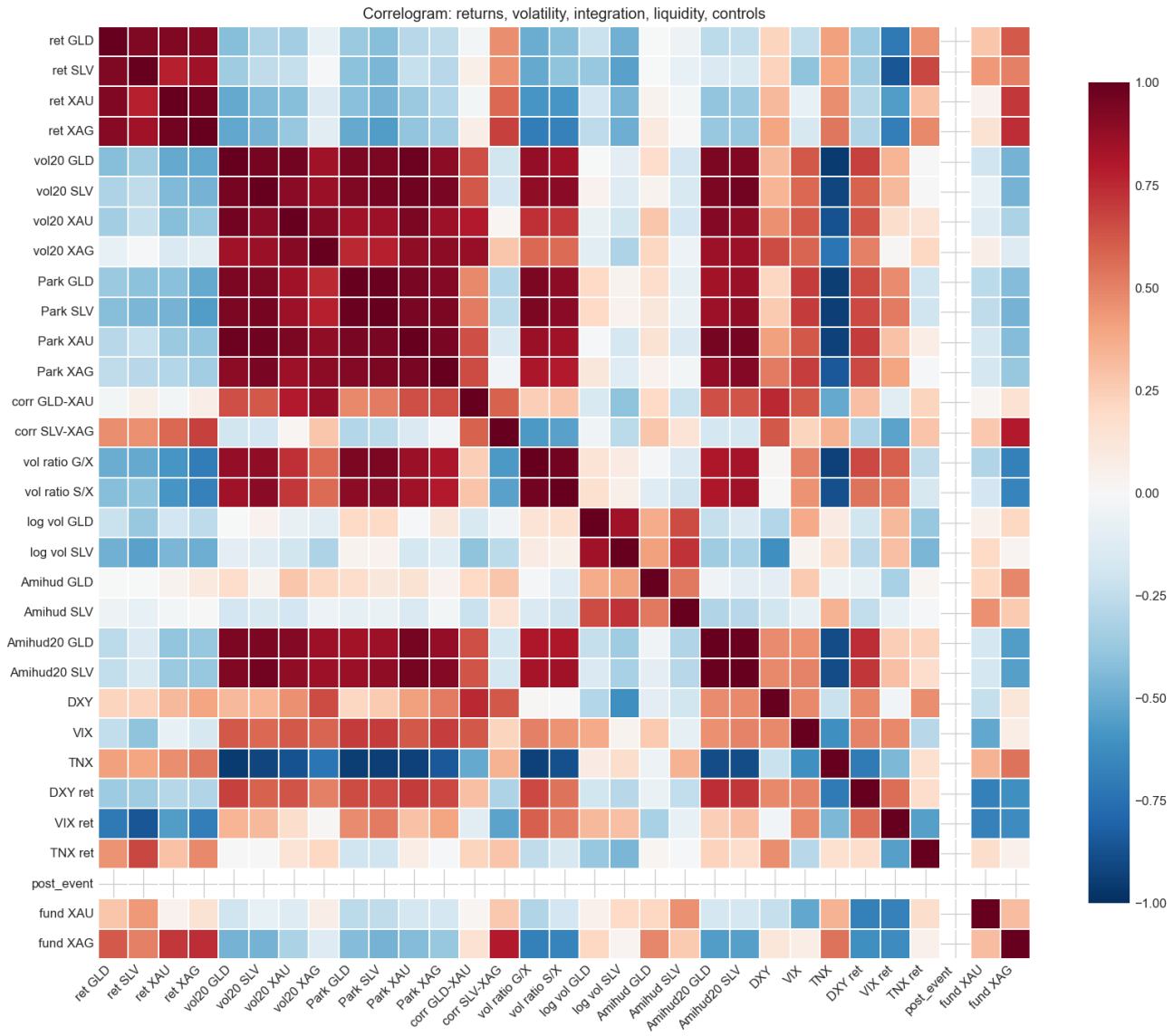
For the most recent 7 days we have *minute-level* data for both ETFs (Yahoo) and Binance. We merge on datetime and compute 1m returns to get a same-frequency measure of short-run co-movement.

```
Minute-level return correlation (last 7 days, overlapping 1m bars):
  GLD-XAU: 0.9913
  SLV-XAG: 0.9934
```

	mean	std	min	max
amihud_gld	7.483538e-10	6.085406e-10	3.474315e-12	2.885031e-09
amihud_20_gld	7.009107e-10	1.577653e-10	4.831240e-10	1.276925e-09
amihud_slv	4.780433e-10	3.276916e-10	1.153311e-12	2.008110e-09
amihud_20_slv	4.779806e-10	1.142224e-10	2.962718e-10	7.452803e-10

3.3 Correlogram

We compute the correlation matrix for a set of variables that are meaningful for the analysis: returns, volatility measures, integration (rolling correlation/beta), liquidity proxies, controls and the event dummy. Only numeric series with sufficient non-missing overlap are included.



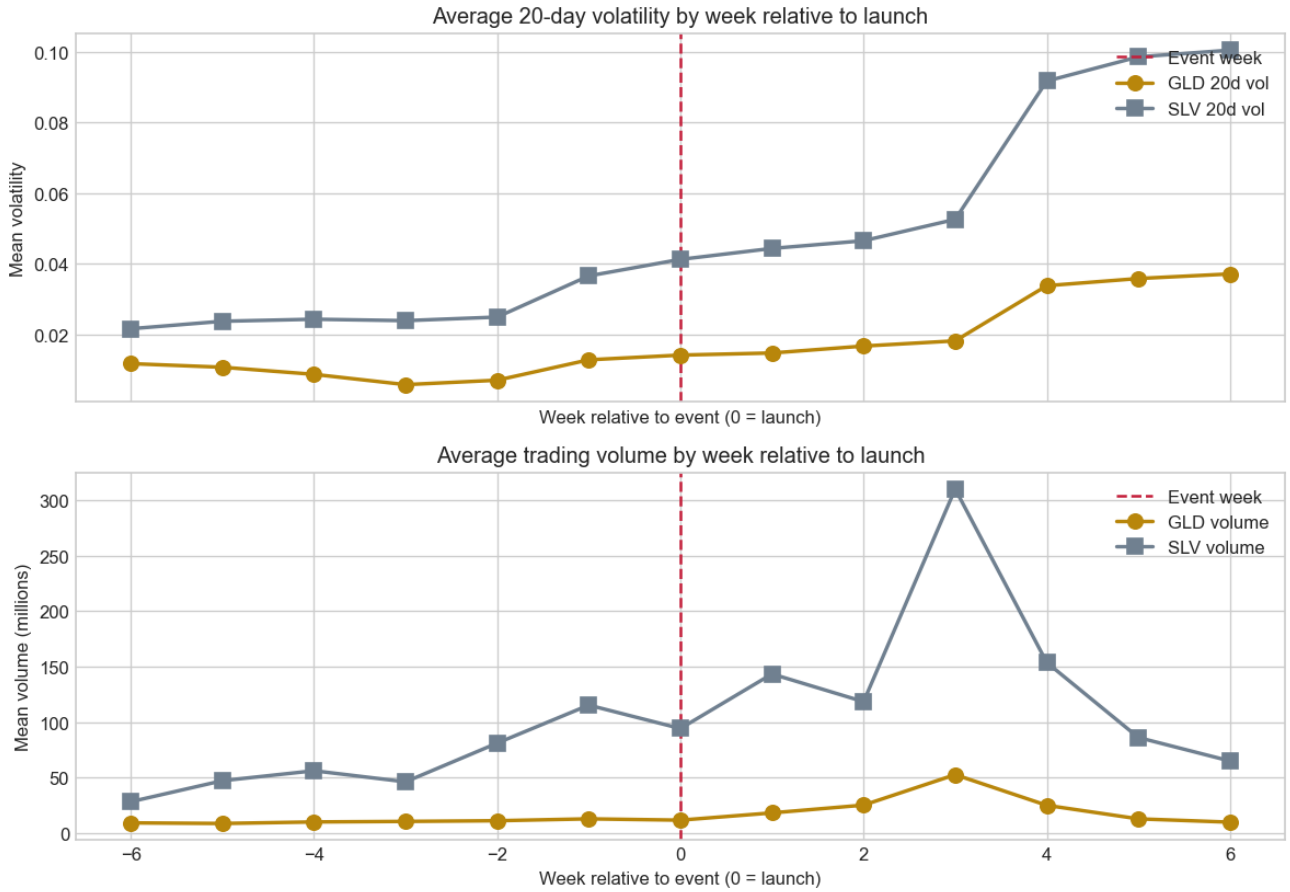
Comment. Variables with the largest absolute correlation with `post_event` are those that move most with the event. This helps spot which outcomes (e.g. volatility, correlation) are most linked to the launch.

4. Event-study style comparison

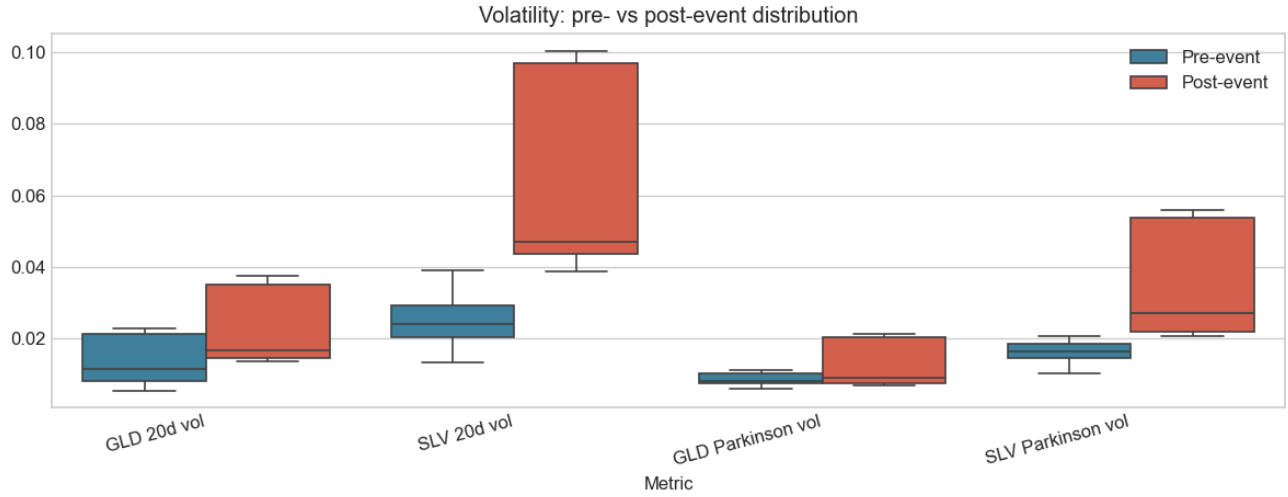
We compare average volatility, volume and integration measures before and after the futures launch. The event date is 2026-01-05 (Binance XAUUSDT TRADFI launch).

Pre vs post: main numbers at a glance. Bar chart of average volatility and volume before and after the event. The jump in the bars after the event is the "effect" we test in regressions.

Dynamics around the event: average outcome by week relative to the launch (week 0 = event week). This shows whether the change is immediate or builds over time.



Pre vs post distributions for key metrics (boxplots by period):



	metric	period	mean	std	n
0	GLD 20d vol	pre	1.343866e-02	6.062648e-03	66
1	GLD 20d vol	post	2.380297e-02	1.011750e-02	32
2	SLV 20d vol	pre	2.411353e-02	6.133201e-03	66
3	SLV 20d vol	post	6.651057e-02	2.643642e-02	32
4	GLD Parkinson vol	pre	8.671084e-03	1.418413e-03	67
5	GLD Parkinson vol	post	1.357468e-02	6.393080e-03	32
6	SLV Parkinson vol	pre	1.625328e-02	2.567734e-03	67
7	SLV Parkinson vol	post	3.669129e-02	1.564470e-02	32
8	GLD volume	pre	1.512372e+07	9.048477e+06	85
9	GLD volume	post	2.289424e+07	1.756732e+07	32
10	SLV volume	pre	4.388922e+07	2.724932e+07	85
11	SLV volume	post	1.439370e+08	9.608540e+07	32
12	corr_20_gld_xau	pre	NaN	NaN	0
13	corr_20_gld_xau	post	9.173261e-01	1.028615e-02	12
14	corr_20_slv_xag	pre	NaN	NaN	0
15	corr_20_slv_xag	post	9.342025e-01	1.902024e-03	9
16	Amihud GLD	pre	6.216244e-10	4.884536e-10	85
17	Amihud GLD	post	1.084979e-09	7.604065e-10	32
18	Amihud SLV	pre	4.955923e-10	3.477594e-10	85
19	Amihud SLV	post	4.314288e-10	2.665129e-10	32
20	GLD GK vol	pre	8.847988e-03	1.160877e-03	67
21	GLD GK vol	post	1.430969e-02	6.899277e-03	32
22	SLV GK vol	pre	1.677668e-02	2.682417e-03	67
23	SLV GK vol	post	3.813931e-02	1.648899e-02	32
24	GLD 10d vol	pre	1.237271e-02	7.157956e-03	76
25	GLD 10d vol	post	2.663677e-02	1.480021e-02	32
26	SLV 10d vol	pre	2.371003e-02	8.887656e-03	76
27	SLV 10d vol	post	7.482800e-02	3.756976e-02	32
28	beta_60_gld_on_xau	pre	NaN	NaN	0
29	beta_60_gld_on_xau	post	NaN	NaN	0
30	beta_60_slv_on_xag	pre	NaN	NaN	0
31	beta_60_slv_on_xag	post	NaN	NaN	0
32	Beta 20d GLD/XAU	pre	NaN	NaN	0
33	Beta 20d GLD/XAU	post	8.634334e-01	4.255616e-02	12
34	Beta 20d SLV/XAG	pre	NaN	NaN	0
35	Beta 20d SLV/XAG	post	8.298093e-01	6.994691e-03	9

Comment. Compare the "pre" and "post" rows for each metric. If the post mean is higher than the pre mean for volatility or correlation, that points to an increase after the event. For volume, higher post means more trading. For liquidity (Amihud), lower post means better liquidity.

For a more formal comparison we can run simple t-tests.

	metric	t_stat	p_value
0	GLD 20d vol	5.347993	3.390943e-06
1	SLV 20d vol	8.955992	2.629522e-10
2	GLD Parkinson vol	4.288773	1.510642e-04
3	SLV Parkinson vol	7.342943	2.473673e-08
4	GLD GK vol	4.448184	9.884403e-05
5	SLV GK vol	7.282962	2.929140e-08
6	GLD 10d vol	5.201790	7.418222e-06
7	SLV 10d vol	7.607690	1.040742e-08
8	GLD volume	2.385871	2.221668e-02
9	SLV volume	5.802936	1.743247e-06
10	Amihud GLD	3.206915	2.601864e-03
11	Amihud SLV	-1.063142	2.912466e-01
12	corr_20_gld_xau	NaN	NaN
13	corr_20_slv_xag	NaN	NaN
14	beta_60_gld_on_xau	NaN	NaN
15	beta_60_slv_on_xag	NaN	NaN
16	Beta 20d GLD/XAU	NaN	NaN
17	Beta 20d SLV/XAG	NaN	NaN

Comment. A low p-value (e.g. below 0.05) means the difference between pre and post is statistically significant. We use **Welch's t-test** (unequal variances, `equal_var=False`) because pre and post variances often differ.

Statistical validation: full t-test table and Levene test

Below we report **Welch t-test** (difference in means) and **Levene test** (difference in variances) for all metrics that have both pre and post data. Significance levels: * 10%, ** 5%, *** 1%.

	Metric	Pre n	Post n	Pre mean	Post mean	t (Welch)	t-test p	Sig.	Levene p
0	GLD 20d vol	66	32	1.343900e-02	2.380300e-02	5.3480	0.0000	***	0.0020
1	SLV 20d vol	66	32	2.411400e-02	6.651100e-02	8.9560	0.0000	***	0.0000
2	GLD Parkinson vol	67	32	8.671000e-03	1.357500e-02	4.2888	0.0002	***	0.0000
3	SLV Parkinson vol	67	32	1.625300e-02	3.669100e-02	7.3429	0.0000	***	0.0000
4	GLD GK vol	67	32	8.848000e-03	1.431000e-02	4.4482	0.0001	***	0.0000
5	SLV GK vol	67	32	1.677700e-02	3.813900e-02	7.2830	0.0000	***	0.0000
6	GLD 10d vol	76	32	1.237300e-02	2.663700e-02	5.2018	0.0000	***	0.0000
7	SLV 10d vol	76	32	2.371000e-02	7.482800e-02	7.6077	0.0000	***	0.0000
8	GLD volume	85	32	1.512372e+07	2.289424e+07	2.3859	0.0222	**	0.0086
9	SLV volume	85	32	4.388922e+07	1.439370e+08	5.8029	0.0000	***	0.0000
10	Amihud GLD	85	32	0.000000e+00	0.000000e+00	3.2069	0.0026	***	0.0067
11	Amihud SLV	85	32	0.000000e+00	0.000000e+00	-1.0631	0.2912		0.4630

Comment. Levene $p < 0.05$ indicates that variances differ between pre and post (e.g. higher volatility in post implies more spread), which supports using Welch's t-test instead of the standard equal-variance t-test.

5. Simple regression models

We estimate simple linear models to quantify the impact of the event and controls on ETF volatility and absolute returns. Following the literature (e.g. Augustin et al.), we winsorize outcome variables at 5% and 95% to reduce

the influence of outliers, and report HAC (Newey–West) standard errors to allow for autocorrelation in daily data.

Date	vol_realized_20_ret_gld	vol_realized_20_ret_slv	dxy_ret	vix_ret	tnx_ret	post_event
2025-09-30	0.007630	0.013341	-0.001431	0.009877	0.001689	0
2025-10-01	0.007610	0.013404	-0.000614	0.000614	-0.010177	0
2025-10-02	0.007558	0.013260	0.001432	0.020657	-0.004393	0
2025-10-03	0.007311	0.013733	-0.001329	0.001202	0.007555	0
2025-10-06	0.007869	0.013770	0.003983	-0.016960	0.010385	0

```
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
const          0.0133      0.002      7.862      0.000       0.010       0.017
post_event     0.0099      0.004      2.458      0.014       0.002       0.018
dxy_ret        0.4099      0.267      1.536      0.125      -0.113       0.933
vix_ret        0.0040      0.007      0.574      0.566      -0.010       0.018
tnx_ret        0.0475      0.219      0.217      0.828      -0.382       0.477
dgs10_ret     -0.0872      0.219     -0.398      0.691      -0.517       0.343
==============================================================================
```

```
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
const          0.0242      0.002     14.149      0.000       0.021       0.028
post_event     0.0410      0.010      4.153      0.000       0.022       0.060
dxy_ret        0.8894      0.681      1.307      0.191      -0.445       2.224
vix_ret       -0.0008      0.015     -0.053      0.958      -0.030       0.029
tnx_ret        0.0597      0.482      0.124      0.901      -0.886       1.005
dgs10_ret     -0.2373      0.517     -0.459      0.646      -1.250       0.775
==============================================================================
```

Comment. The `post_event` coefficient is the main result: it shows how much volatility changed after the launch, once we control for DXY, VIX and 10Y yield changes. If it is positive and the p-value (P>|t|) is below 0.05, we conclude that volatility increased after the event (H1 supported for that ETF).

If needed, we can also model absolute returns as a simple proxy for volatility:

```
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
const          0.0091      0.001      7.575      0.000       0.007       0.012
post_event     0.0099      0.003      3.343      0.001       0.004       0.016
dxy_ret        0.0588      0.330      0.178      0.859      -0.588       0.705
vix_ret        0.0043      0.012      0.361      0.718      -0.019       0.028
tnx_ret        0.2460      0.312      0.789      0.430      -0.365       0.857
dgs10_ret     -0.2769      0.277     -1.000      0.318      -0.820       0.266
==============================================================================
```

```
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
const          0.0206      0.003      7.928      0.000       0.016       0.026
post_event     0.0246      0.004      6.232      0.000       0.017       0.032
dxy_ret       -0.1863      0.462     -0.403      0.687      -1.092       0.719
```

```
vix_ret        0.0137      0.020      0.668      0.504      -0.026       0.054
tnx_ret        1.0722      0.453      2.368      0.018       0.185       1.960
dgs10_ret     -0.8386      0.500     -1.677      0.094      -1.819       0.142
==============================================================================
```

Liquidity (H2): we also regress Amihud illiquidity and log(volume) on the event and controls. A negative coefficient on `post_event` for Amihud would indicate improved liquidity (lower illiquidity) after the launch; a positive coefficient for log_vol would indicate higher volume.

```
Amihud GLD:
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
const       6.156e-10   5.09e-11     12.105      0.000    5.16e-10    7.15e-10
post_event   4.42e-10   1.93e-10      2.290      0.022    6.37e-11     8.2e-10
dxy_ret    -8.505e-09   1.66e-08     -0.513      0.608    -4.1e-08     2.4e-08
vix_ret     1.167e-10   5.95e-10      0.196      0.844   -1.05e-09    1.28e-09
tnx_ret    -7.687e-10    1.8e-08     -0.043      0.966   -3.61e-08    3.45e-08
dgs10_ret   1.339e-09    1.7e-08      0.079      0.937   -3.19e-08    3.46e-08
==============================================================================
Amihud SLV:
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
const       4.643e-10   2.62e-11     17.705      0.000    4.13e-10    5.16e-10
post_event -1.917e-11   7.92e-11     -0.242      0.809   -1.74e-10    1.36e-10
dxy_ret      2.54e-09   7.19e-09      0.353      0.724   -1.15e-08    1.66e-08
vix_ret    -2.658e-11   2.62e-10     -0.102      0.919    -5.4e-10    4.86e-10
tnx_ret      1.82e-08   7.59e-09      2.399      0.016    3.33e-09    3.31e-08
dgs10_ret  -1.345e-08      8e-09     -1.681      0.093   -2.91e-08    2.23e-09
==============================================================================
Log volume GLD:
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
const         16.4061      0.089    185.158      0.000      16.232      16.580
post_event     0.3168      0.194      1.630      0.103      -0.064       0.698
dxy_ret       11.5705     13.162      0.879      0.379     -14.226      37.367
vix_ret        0.2885      0.488      0.591      0.555      -0.669       1.246
tnx_ret        9.9547     12.454      0.799      0.424     -14.454      34.364
dgs10_ret    -13.7500     11.573     -1.188      0.235     -36.433       8.933
==============================================================================
Log volume SLV:
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
const         17.4529      0.113    153.861      0.000      17.231      17.675
post_event     1.1273      0.161      7.004      0.000       0.812       1.443
dxy_ret        5.1501     11.921      0.432      0.666     -18.214      28.515
vix_ret        0.5086      0.521      0.975      0.329      -0.513       1.531
tnx_ret       26.8226     12.569      2.134      0.033       2.189      51.457
dgs10_ret    -24.9975     13.407     -1.865      0.062     -51.274       1.279
==============================================================================
```

6. Interpretation and next steps

6.1 Key results summary

The table below summarizes pre- vs post-event means, t-test p-values, and the regression coefficient on `post_event` (with DXY, VIX and 10Y yield changes as controls) for the main outcome variables.

	Metric	Pre mean	Post mean	t-test p	post coef	reg p
0	GLD 20d vol	0.0134	0.0238	0.0	0.0099	0.014
1	SLV 20d vol	0.0241	0.0665	0.0	0.041	0.0
2	GLD Parkinson vol	0.0087	0.0136	0.0002	0.0046	0.057
3	SLV Parkinson vol	0.0163	0.0367	0.0	0.0195	0.001
4	GLD GK vol	0.0088	0.0143	0.0001	0.0051	0.048
5	SLV GK vol	0.0168	0.0381	0.0	0.0206	0.001
6	GLD 10d vol	0.0124	0.0266	0.0	0.0141	0.013
7	SLV 10d vol	0.0237	0.0748	0.0	0.0497	0.0
8	GLD volume	15.12M	22.89M	0.0222	5.99e+06	0.134
9	SLV volume	43.89M	143.94M	0.0	8.03e+07	0.0
10	Amihud GLD	0.0	0.0	0.0026	4.42e-10	0.022
11	Amihud SLV	0.0	0.0	0.2912	-1.92e-11	0.809
12	Corr GLD–XAU	—	0.9173	N/A	N/A	N/A
13	Corr SLV–XAG	—	0.9342	N/A	N/A	N/A
14	Beta 60d GLD/XAU	—	—	N/A	N/A	N/A
15	Beta 60d SLV/XAG	—	—	N/A	N/A	N/A
16	Beta 20d GLD/XAU	—	0.8634	N/A	N/A	N/A
17	Beta 20d SLV/XAG	—	0.8298	N/A	N/A	N/A

Comment. This table brings everything together. Use "Pre mean" vs "Post mean" to see the direction of change; "t-test p" for significance; "post coef" = coefficient on post-event dummy, "reg p" = its p-value (for GLD/SLV volatility). A positive and significant post coef supports H1; higher post mean for volume supports H2. Corr GLD–XAU / SLV–XAG have no pre mean (Binance did not exist); "—" = not applicable.

6.2 Empirical results

```
Volatility (H1): After controlling for DXY, VIX and 10Y yield changes, the post-event coefficient in the

For SLV, the post_event coefficient is 0.04105 (p = 0.000): volatility increased and is significant.

Integration (descriptive only): Average rolling correlation GLD-XAU is 0.917 post-event; SLV-
XAG is 0.934 post-event. There is no pre-event correlation (Binance did not exist), so we do not test a p

Liquidity (H2): Average GLD volume is 22894241 post-event vs 15123725 pre-event; SLV volume can be read f
```

6.3 Conclusions on hypotheses

Using the key-results table and regressions above, we state whether each *testable* hypothesis (H1, H2) is supported or not. Integration is not tested as a hypothesis (no pre-period); we only describe post-event correlation.

```
H1 (volatility increase after launch):
  GLD: supported (positive and significant post_event means higher volatility).
  SLV: supported.

Integration (descriptive, post-event only; no pre/post test):
  Rolling correlation GLD-XAU and SLV-XAG are reported in section 3.2 and the key-results table.
  We do not test a formal integration hypothesis because Binance data do not exist before the launch.
```

```
H2 (improved liquidity: higher volume, lower Amihud):
  GLD volume: higher post (higher post supports H2).
  SLV volume: higher post.
  Check Amihud in the pre/post table; lower post = better liquidity.
```

6.4 Limitations and extensions

- **No pre-period for integration.** Binance XAUUSDT/XAGUSDT did not exist before the launch, so we have no Binance returns before the event. Rolling correlation (GLD–XAU, SLV–XAG) and beta can only be computed from the launch date onward. We cannot compare "integration before vs after" in the same way as for volatility or liquidity; we can only describe the level and evolution of integration post-launch.
- **Short post-event window.** We have limited days after the Binance launch. Results can change when more data are available.
- **Robustness.** Try different event dates, volatility windows (e.g. 10d or 60d), or placebo tests (fake event date) to see if the effects hold.
- **Beta.** The key-results table focuses on volatility, volume and correlation. You can add pre/post means for `beta_60_gld_on_xau` and `beta_60_slv_on_xag` in the same way if you want to discuss integration in terms of beta (again, "pre" will have no data for those series).

6.5 Conclusion

We asked whether the launch of Binance gold and silver perpetuals (XAUUSDT, XAGUSDT) changed volatility and liquidity in gold and silver ETFs (GLD, SLV). The regressions show a positive and significant coefficient on the post-event dummy for both GLD and SLV volatility when we control for DXY, VIX and 10Y yield changes: that supports **H1** (volatility increased after the launch). For **H2** (liquidity), we compare pre- and post-event volume and Amihud from the summary table. We do not test a formal hypothesis on integration (no pre-event Binance data); we only report post-event rolling correlation as descriptive evidence. In short: the evidence points to higher ETF volatility after the Binance TRADFI launch; liquidity and integration are discussed in the tables and section 6.3. Because the post-event window is still short, updating the analysis with more data is recommended.